Scalability & Future Plan

Date	06-07-2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	1 Mark

Scalability & Future Plan:

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Application currently runs on a local system using Streamlit.	Limited accessibility for multiple users.	High
2	User interaction history is stored in CSV files.	Not suitable for handling large volumes of data.	Medium
3	Internet connection is required for Google Gemini API responses.	Not suitable for handling large volumes of data.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports local users.	Deploy the application on cloud platforms such as Google Cloud Run or Streamlit Community Cloud to support multiple concurrent users.
2	Data Storage	Uses CSV files to store user sessions and prediction history.	Replace CSV storage with a scalable database such as MySQL, PostgreSQL, or Firebase.
3	Performance	Emotion detection is performed locally using BiLSTM and BERT models.	Optimize model loading, implement caching, and utilize GPU acceleration for faster predictions
4	Security	Uses local authentication and a .env file for API key management.	Implement secure user authentication, encrypted databases, HTTPS, and role-based access control.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud deployment for public access.	Next 3 Months	Enables multiple users to access the application online.
Phase 2	Multilingual emotion detection and recommendations.	Next 6 Months	Makes the application accessible to users from different language backgrounds.
Phase 3	Voice-based emotion detection and speech analysis.	Next 9 Months	Improves accessibility and provides more natural user interaction.
Phase 4	Integration with Learning Management Systems (LMS) and personalized learning progress tracking.	Next 12 Months	Provides a comprehensive AI-powered learning support platform with long-term student progress monitoring.